

Flatcar OS Configuration

Source section: `os-config/` (renamed from `setup/` in the refactored branch, with security split out into its own `security/` section). Covers host configuration, networking, storage, and systemd.

Host Configuration (`os-config/host-config/`)

Topic	How
Adding users	Butane <code>passwd.users</code> ; default user is <code>core</code> (<code>wheel/sudo</code>)
Date & timezone	<code>timedatectl set-timezone</code> ; NTP via <code>systemd-timesyncd</code>
Kernel modules / sysctl	<code>/etc/modules-load.d/</code> and <code>/etc/sysctl.d/</code>
Drop-in units	<code>/etc/systemd/system/NAME.d/</code> override files
Environment variables	<code>EnvironmentFile=</code> or <code>Environment=</code> in unit files
Timers (cron replacement)	<code>.timer</code> + <code>.service</code> unit pairs
Power management	systemd power targets; Tuned profiles
NVIDIA GPUs	<code>nvidia-drivers-*</code> sysext (available since 4344.0.0)

Setting update strategy via Butane (see also [flatcar-updates](#)):

```
variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/flatcar/update.conf
      overwrite: true
      mode: 0420
      contents:
        inline: |
          REBOOT_STRATEGY=etcd-lock
```

Networking (`os-config/network/`)

Flatcar uses `systemd-networkd` for all network configuration.

Custom network config via Butane:

```

variant: flatcar
version: 1.0.0
storage:
  files:
    - path: /etc/systemd/network/10-static.network
      contents:
        inline: |
          [Match]
          Name=eth0
          [Network]
          Address=192.168.1.100/24
          Gateway=192.168.1.1
          DNS=8.8.8.8

```

Topic	File / Tool
DNS	/etc/systemd/resolved.conf or .network [Network]DNS=
ACPI	/etc/acpi/ custom tables (os-config/network/ACPI.md)
OverlayBD	Artifact streaming sysex (overlaybd , since 4426.0.0)

Storage (os-config/storage/)

Topic	Notes
RAID	Software RAID via mdadm ; recommended for root filesystem
Adding disk space	Partition and format via Ignition or runtime fdisk / mkfs
Mounting storage	Systemd .mount units or /etc/fstab
iSCSI	open-iscsi included in base image
ZFS	Opt-in sysex (zfs) since 3913.0.0
LVM	lvm2 tools included
Swap	Swap file or partition via Butane storage section

RAID example (Butane):

```

variant: flatcar
version: 1.0.0
storage:
  raid:
    - name: md0

```

```

    level: raid1
    devices:
      - /dev/sdb
      - /dev/sdc
  filesystems:
    - device: /dev/md/md0
      format: ext4
      path: /var/lib/data
      wipe_filesystem: true

```

systemd (os-config/systemd/)

Topic	How
Getting started	systemctl basics; journalctl for logs
Drop-in units	systemctl edit SERVICE or files in .d/ directories
udev rules	/etc/udev/rules.d/ — hardware event automation
User units	~/.config/systemd/user/ for per-user services
systemctl	Full systemctl reference for Flatcar

Managing a Docker container as a systemd service:

```

[Unit]
Description=My Application
After=docker.service
Requires=docker.service

[Service]
ExecStartPre=-/usr/bin/docker rm --force myapp
ExecStart=/usr/bin/docker run --name myapp --rm myimage:latest
Restart=always
RestartSec=5s

[Install]
WantedBy=multi-user.target

```

Place this in Butane under `systemd.units` with `enabled: true`.